

Vascularization reconstruction using Contrast-Enhanced Micro CT for high-resolution X-ray based 3D histology

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Preface

Biblio using Zotero

Report using Typst

Report template: <https://github.com/benjaminееkh/kul-typst-template>

Report code: <https://github.com/Nachtraven/ucl-2026-based-on-kul-typst-template/tree/main>

Thesis content code: <https://github.com/Nachtraven/Thesis-Sean-Nachtrab-EPL-25-26>

This is a preface - TO BE COMPLETED

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Abstract

The accurate reconstruction of tissue vascularization in ex-vivo biopsies is essential to quantitative analysis of samples, to enable their comparison across various treatment profiles, and achieve the level of accuracy required for drug research. We aim to enable the analysis of antiangiogenic drugs for cancer treatment, specifically focused on pazopanib, by improving quantification of vascularization. Classical histology methods such as 2D histology suffer from a lack of accuracy due to the slicing process, and when used for vascularization reconstruction, lose the ability to reconstruct its structural nature. 3D techniques such as contrast enhanced Micro CT enables the acquisition of high resolution 3D images of ex-vivo samples aided by a contrast agent, but does not enable direct quantitative information extraction due to discontinuities, noise, low contrast and small features. In this document, a method for the structural reconstruction of vascularization from microCT images is presented, enabling analysis of the impact of antiangiogenic drugs on the vascularization of tumors in mice.

-> this can be improved and extended later

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Chapter 1

Introduction

The analysis of soft tissue is foundational to modern medicine, being used everywhere from clinical environments to laboratories. Tissue samples are collected and imaged in order to gain an understanding of their structure and composition, called histology or histopathology in the case of diseased tissue. We use the term histology to refer to either interchangeably. Classically, this sampling and subsequent analysis has been done in 2D, with the microscope being the catalyst for the development of the field, and being the gold standard for this process. However, this form of histology presents multiple limitations that stem from its 2D nature: the loss of 3D understanding and structure, as well as very limited quantification capabilities. [0.1]

Histology is relevant in both clinical and laboratory environments. Clinically, the goal of a biopsy and subsequent analysis under the microscope is generally to identify a qualitative parameter in order to enable diagnosis and treatment: identifying the presence of a kind of cell, the phenotyping of a cell, or the estimation of a parameter, for example in cancerology the presence or absence of blood vessels in a tumor. [0.2]

Such biopsies done in 2D present the distinct disadvantage of being unable to obtain information at the same granularity in all three axis: microscope slides in 2D may be stacked to construct a 3D understanding, but these stacked planes are separated by a dead space where no information is acquired [0.3]. This loss of accuracy is catastrophic for the reconstruction and quantification of structures that are small and where their structure is relevant to their function, such as blood vessels, on top of the deformation caused by the cutting process.

Blood vessel understanding and reconstruction has emerged over the past years as a critical element in understanding and pushing the boundaries in new drug research: angiogenesis, or the creation of new blood vessels, is a key step in the growth and spread of cancerous tumors. Classical methods, relevant for qualitative understanding and estimation, fall short in terms of accuracy for quantitative analysis for clinical trials [0.4]. 2D reconstruction suffers due to its inherent variability resulting from the slicing process and lack of resolution, with the field is progressing towards the use of high resolution 3D data for these quantitative analysis tasks.

We will begin by diving into the current state of the art in histology for drug screening,

followed by the MicroCT methods used, and finish with an exploration of the methods of image understanding more broadly,

Should this be extended with the aetiology of cancer and angiogenesis?

- 0.1 Appears interesting: Limitations of clinical and biological histology <https://doi.org/10.1054/mehy.1999.0894>
- 0.2 They speak of “presence of vascular or stromal invasion”: Histology: The gold standard for diagnosis? <https://pubmed.ncbi.nlm.nih.gov/37008634/>
- 0.3 a. 3D reconstruction from 2D slices “Extending two-dimensional histology into the third dimension through conventional micro computed tomography” <https://doi.org/10.1016/j.neuroimage.2016.06.005>
- 0.3 b. Other source for 3D reconstruction from 2D slices “A Survey of Methods for 3D Histology Reconstruction” <https://doi.org/10.1016/j.media.2018.02.004>
- 0.4 Sources on angiogenesis in tumors from Wlodarski

Chapter 2

State of the art

State of the art from “Scientific writing-tips and tricks_Deadlines.pdf”:

In the state of the art, you indicate what has been studied, why it has been studied, and, in general terms, how it has been studied.

This section contains:

- ✓ The background of the study (= context);
- ✓ Research that has been done in the frame of your thesis topic;
- ✓ Information that is needed for the reader to understand your topic and the remaining scientific issues/problems.

➔ Make a fluent story of it, and not just a sum up of different papers or references.

➔ The context (per paragraph or section) usually ends with a conclusion or problem statement that sets the scene for your specific research questions and project aim + objectives.

With the goal of reconstructing tumor vascularization for the analysis of the antiangiogenic drug Pazopanib in the context of cancer treatment, we begin by exploring the use of histology for drug screening, followed by imaging methods, more specifically contrast-enhanced Micro CT, and finish with an exploration of the methods of structure extraction from images.

2.1 Histology for drug screening

In order to bring new drugs to market, a set of steps must be followed that are established by organisations such as the European Medical Association in the European Union, or the Food and Drug Administration in the U.S [1.1]. One of the key steps is collecting data on the effectiveness of the drug, done through clinical trials. These trials, before taking place in humans, are done on animals in a controlled lab environment, where collected data is analyzed for the outcomes and side effects of the drug. In the analysis process for Pazopanib, tumors are sampled and analysed ex-vivo, the histology of these tumors is critical in identifying the impact of treatment when compared to a control.

The conclusions taken from each step in the clinical trial decide if the trial moves on, meaning that a more accurate prediction with tighter confidence intervals enables a better discriminative power and reduces the amount of effective drugs discarded due to uncertainty from process limitations.

2.1.1 Histology in industry

Histology is in use in industry for clinical trials amongst other tasks, and the field is growing [2.1]. Histology is used due to its large amounts of qualitative data and high discriminative power, and is interpreted by experts during the process of drug research. It is used in clinical trials of structure altering drugs, or where the effect of interest is expected to be visible, and is generally done by specialized histopathologists [2.1]. Clinical trials vary widely in their design, as well as have a large variety of possible targets, in oncology specifically, 2D histology remains the most widely used technique [2.3], [2.27, 2.60] for the histology of human biopsies. For the analysis of Pazopanib, 2D histology is the reference method used to evaluate tumors.

2D histology

The field of histology is born from 2D histology of tissue under microscope magnification. 2D microscope slice histology, known as “classical histology”, takes the form of tissue sample collection, often embedding of the tissue, followed by staining with various agents, applied to increase contrast or highlight certain structures, and ending in investigation using optical or electron microscopy [3.1]. This technique is used due to its delivery of a large amount of relevant quantitative data immediately interpretable by experts, and high discriminative power, although it is time consuming and is costly as a result [3.2]. As a technique it has evolved in multiple directions, from the utilization of advanced staining agents [3.2] allowing the staining of a broad range of classes (DNA, proteins, lipids, or carbohydrates), to different lighting wavelengths, ranging from ultraviolet [3.3] to infrared [3.4].

2D histology is limited by its requirement to slice the target tissue before analysis, resulting in destructive modification of the sample. This slicing has the potential of deforming the

tissue, with various techniques developed to counteract this [3.5] such as embedding in a wide variety of mediums. Slicing is also not orientation agnostic, and a large number of slices might be required for a sufficient spatial analysis of the sample, especially in situations where the observation target does not follow a particular axis, as with vascularization.

The limitations of pure 2D histology are now resulting in quantitative histology moving in the direction of 3D analysis, especially when 3D structure of the tissue is relevant.

Unmentioned above:

- samples dry out in classical histo,
- the resolution is limited when stacking
- there are methods for extracting some 3D info from the 2D slices ex: “confocal microscopy light sheet microscopy or optical coherence tomography” <https://doi.org/10.1038/s41467-022-34048-4>

3D histology

3D histology is an evolution of 2D histology, increasing the amount of information available for analysis, and as a result improving discriminative power when compared to 2D classical histology. In its most basic form as an evolution of classical 2D histology, it involves taking multiple 2D slices and aligning then stacking them to create a virtual third axis [4.1]. 3D histology is relevant for tumor analysis [4.2], allowing for structural understanding of a tissue, such as in [4.2] enabling “the spread and infiltration of invasive carcinoma to be understood”. This form of histology is often still done in the same methods as with 2D, where a human carries out interpretation, requiring common standards and communication, such as in [4.3]

Other well known methods for 3D histology that do not require slicing and thus avoid sample damage and destruction are magnetic resonance imaging (MRI) and computed tomography (CT), each having a higher resolution variant, micro-MRI and microCT respectively. These techniques enable virtual slicing, allowing observation across any plane, and have resolutions of tens microns for micro-MRI down to sub-micrometer scale for MicroCT. MRI is interesting for tissue observation due to its high contrast on soft tissue, as opposed to CT that performs best on calcified tissue. Both methods however are able to make use of contrast-enhancing staining agents (CESAs) [4.4,4.5] resulting in the ability for microCT to be readily used for imaging soft tissue, and is termed contrast-enhanced microCT (CECT) and reviewed for ex-vivo data acquisition in [4.6]. CECT is of particular interest for ex-vivo 3D histology due to the wide variety of staining agents available, and has proven its use in canine heart analysis [4.7] and for vasculature exploration of small animals [4.8].

Not certain about if or how to place the quote, as it is specifically for cancer.

2.1.1.2.1 CECT Imaging

Full section on the state of the art of CECT - TODO

2.2 Structure reconstruction (or: Vascularization reconstruction, Interpretation methods of 3D imaging?)

This section is leaning towards segmentation, as in survey [5.1]

With the goal of reconstructing vascularization in imaged tissue, we aim to take in data from an imaging modality, process it through a software pipeline, and obtain at output a data structure with higher information density than the input. This process of reducing the amount of raw data, but increasing the utility or information content of the data, is explored widely in the field of computer science. It underlies concepts such model fitting to noisy data [5.2] or as more commonly carried out in a biological context and industry, image segmentation, where the goal is to separate regions of a 2D image into multiple segments and objects [5.3].

2.2.1 Segmentation

Image segmentation plays a central role in medical image analysis by enabling quantitative analysis of different kinds: by classifying pixels with semantic labels (semantic segmentation), partitioning of objects (instance segmentation) or a combination of both across the entire image space (panoptic segmentation) [5.1]. The task can be as simple and as old as separating objects from a background, a problem that has been explored in computer science for multiple decades [5.4], or more complex as in panoptic segmentation, only being posed in 2018 [5.5].

Methods of Segmentation

Segmentation methods have evolved over time: from classical computer vision to machine learning and later deep learning methods

TODO: Extend history to highlight the progression in complexity and the higher and higher levels of information extraction achieved

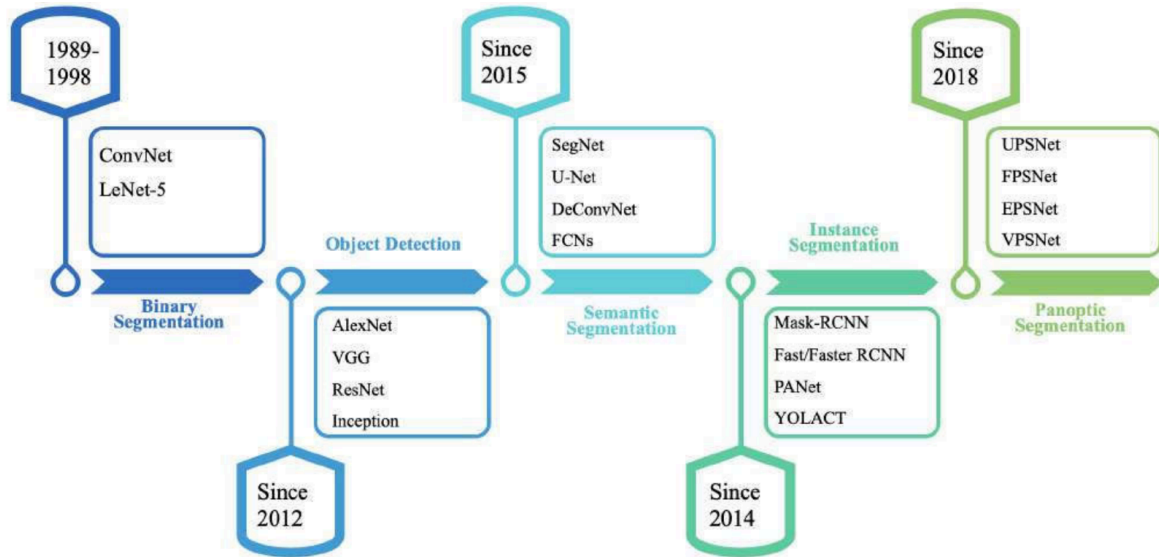


Figure 2.1: From (6.1) Timeline evolution of image segmentation (better version to be found or created)

Imaging and Segmentation of vasculature

Vasculature reconstruction and segmentation of 3D images often requires human segmentation [6.2]. Alternative automated methods are less precise

Sources to be investigated and read:

1. 2013 Application of Micro-Computed Tomography With Iodine Staining to Cardiac Imaging, Segmentation, and Computational Model Development
2. 2010 Micro computed tomography for vascular exploration
3. 2025 Micro-computed tomography to visualize preserved vascular architecture in decellularized human vaginal tissue: explorative study
4. 2004 Micro-computed tomography of the vasculature in parenchymal organs and lung alveoli

Non tomo:

1. 2021 Robust segmentation of vascular network using deeply cascaded AReN-UNet
2. 2023 Vessel Delineation Using U-Net: A Sparse Labeled Deep Learning Approach for Semantic Segmentation of Histological Images
3. 2015 Quantification of Microvascular Tortuosity during Tumor Evolution Using Acoustic Angiography

Vasc references:

1. 1976 The Vascularization of Tumors
2. 2013 Springer Nature, "Vascularization"
3. 1982 Vascularization of Tumors: A Review

Sources, will be re-done once finalized:

- 1.1 <https://www.ema.europa.eu/en/human-regulatory-overview/research-development>
- 2.1 <https://doi.org/10.1111/his.14099>
- 2.2 Not great, behind paywall “Challenges Faced by Cross-sectional Imaging and Histological Endpoints in Clinical Trials” <https://doi.org/10.1093/ecco-jcc/jjw161>
- 2.3 not the best source “The dream and reality of histology agnostic cancer clinical trials” <https://doi.org/10.1016/j.molonc.2014.06.002>
- 2.27 Wlodarski 27: <https://pubmed.ncbi.nlm.nih.gov/32445458/>
- 2.60 Wlodarski 60: <https://pubmed.ncbi.nlm.nih.gov/37008634/>
- 3.0 Section partially inspired from <https://doi.org/10.1155/2019/8617406>
- 3.1 Source from [0], Mescher, L. A. Junqueira’s basic histology. Text and atlas 14th ed. (McGraw-Hill, 2016).
- 3.2 <https://www.ncbi.nlm.nih.gov/books/NBK557663/>
- 3.3 <https://pmc.ncbi.nlm.nih.gov/articles/PMC6223324/>
- 3.4 <https://pmc.ncbi.nlm.nih.gov/articles/PMC10408309/>
- 3.5 Source from [3.0]: <https://doi.org/10.1155/2019/8617406>
- 4.1 <https://doi.org/10.1016/j.ajpath.2012.01.033>
- 4.2 <https://doi.org/10.1136/jcp.2004.024794>
- 4.3 Veterenary based? Not strongest article: Tseng LJ, Matsuyama A, MacDonald-Dickinson V. Histology: The gold standard for diagnosis? Can Vet J. 2023 Apr;64(4):389-391. PMID: 37008634; PMCID: PMC10031787
- 4.4 Single Ho 3+-doped upconversion nanoparticles for high-performance T2-weighted brain tumor diagnosis and MR/UCL/CT multimodal imaging: <https://doi.org/10.1002/adfm.201401609>
- 4.5 Three-dimensional non-destructive soft-tissue visualization with X-ray staining microtomography: <https://doi.org/10.1038/srep14088>
- 4.6 Review Greet: Contrast-Enhanced MicroCT for Virtual 3D Anatomical Pathology of Biological Tissues: A Literature Review <https://doi.org/10.1155/2019/8617406>
- 4.7 Application of Micro-Computed Tomography With Iodine Staining to Cardiac Imaging, Segmentation, and Computational Model Development <https://doi.org/10.1109/tmi.2012.2209183>
- 4.8 Micro computed tomography for vascular exploration <https://doi.org/10.1186/2040-2384-2-7>
- 5.1 IEEE Image Segmentation Using Deep Learning: A Survey <https://doi.org/10.1109/TPAMI.2021.3059968>
- 5.2 RANSAC <https://doi.org/10.1145/358669.358692>
- 5.3 FIRST EDITION! Computer Vision: Algorithms and Applications. Berlin, Germany
- 5.4 OTSU <https://doi.org/10.1109/TSMC.1979.4310076>
- 5.5 Panoptic <https://doi.org/10.48550/arXiv.1801.00868>
- 5.6 U-Net <https://doi.org/10.48550/arXiv.1505.04597>
- 6.1 Panoptic Segmentation: A Review <https://doi.org/10.48550/arXiv.2111.10250>
- 6.2 pCT visualize preserved vascular architecture in decellularized human vaginal tissue: explorative study <https://doi.org/10.1038/s41598-025-14452-8>

2.3 Problem statement, aim and objectives

From (Scientific writing-tips and tricks_Deadlines.pdf):

- ✓ Formulate again relevant, simple, measurable, and feasible research questions that are still remaining (should be clear after reading the SOTA)
- ✓ State the research aim or hypothesis of the project, and formulate concrete objectives

2.3.1 Connected papers

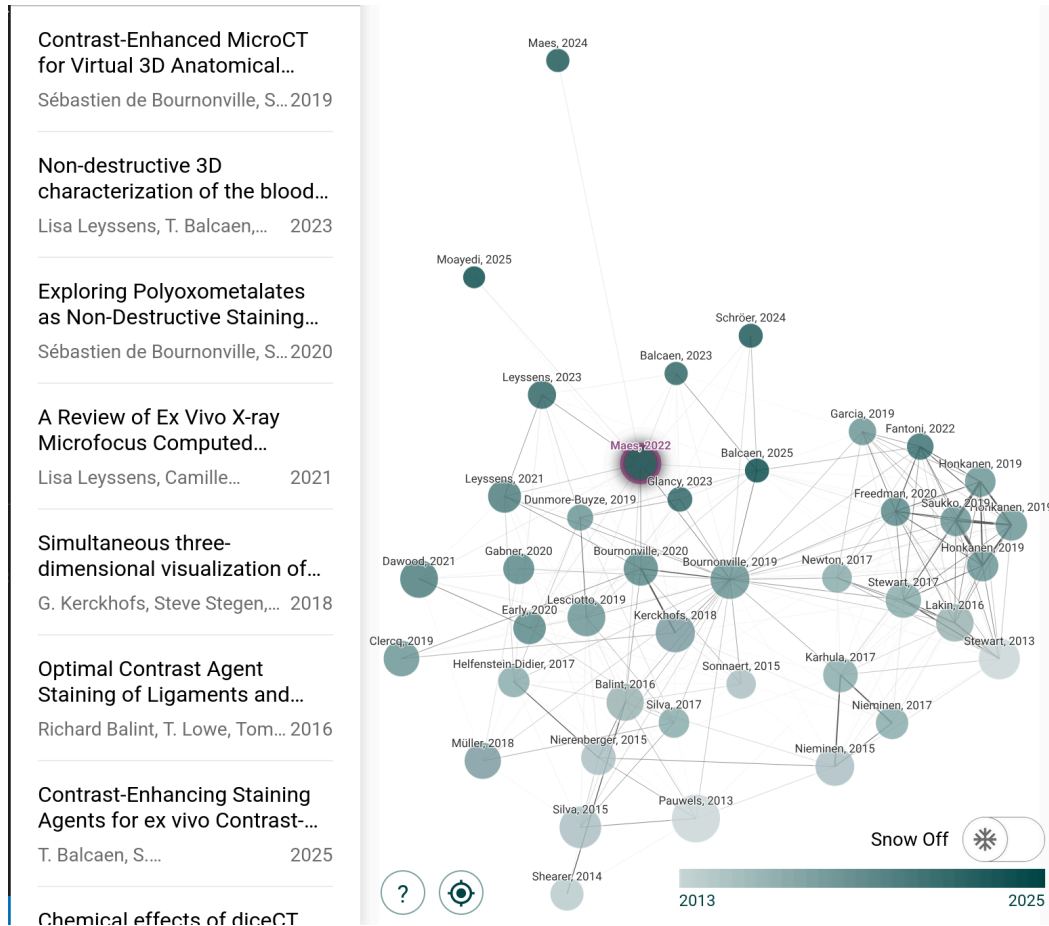


Figure 2.2: Connected papers of Cryogenic contrast-enhanced microCT enables nondestructive 3D quantitative histopathology of soft biological tissues

Bibliography